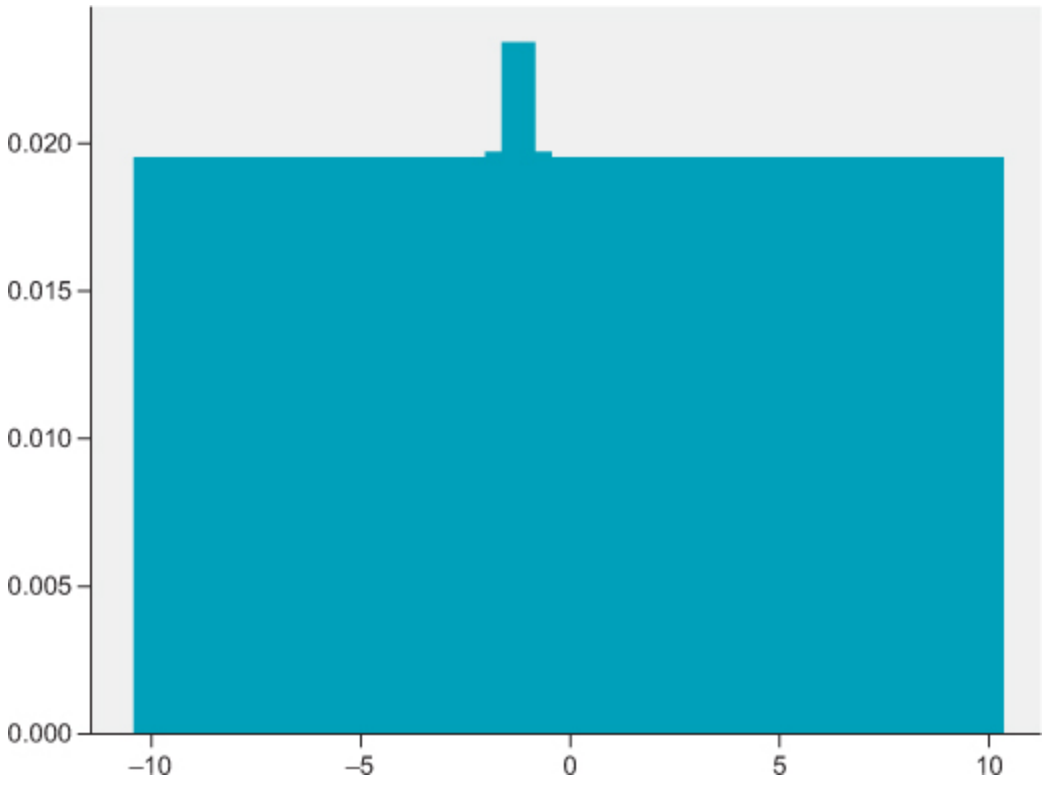


Now let’s see how the distribution changes when we observe a sequence of varying rewards. (We have just made up this sequence of rewards; they do not come from the Freeway game.) We should be able to observe multimodality.

Listing 7.4. Redistributing probability mass with a sequence of observations

```
ob_rewards = [10,10,10,0,1,0,-10,-10,10,10]
for i in range(len(ob_rewards)):
    Z = update_dist(ob_rewards[i],
torch.from_numpy(support).float(), Z, \
                    lim=(vmin,vmax), gamma=0.5)
plt.bar(support, Z)
```

You can see in figure 7.13 that there are now four peaks of varying heights corresponding to the four different kinds of rewards observed, namely 10, 0, 1, and –10. The highest peak (mode of the distribution) corresponds to 10, since that was the most frequently observed reward.

Figure 7.13. This is the result of updating an initially uniform probability distribution after observing a sequence of different rewards. Each “peak” in the distribution corresponds to an observed reward.